Microplastic Size Distributions and Concentration Methods in Freshwater

Evidence from Rivers and Ponds, Semenyih River, and University of Nottingham Malaysia

Year 3 Group Project | Reproducible corpus-based literature screening

30 October 2026

Abstract

This reproducible evidence-map review addresses three connected questions: (1) which microplastic (MP) size fractions are reported as common in river and pond water, accounting for measurement cutoffs, (2) what the Semenyih River literature actually establishes about MP abundance, morphology and size, and (3) which UNM-affiliated studies document methods, instruments, validation and findings relevant to river-water MP concentration. A reproducible title/abstract first pass identified 2,085 potential freshwater-size records, of which 17 were manually curated as transparent examples of reported size evidence. Separately, 33 records mentioned a Nottingham institution; nine explicitly matched the Malaysia campus, while 24 did not satisfy the Malaysia-campus rule. The Semenyih-specific field paper reports abundance and morphology, but its accessible abstract does *not* establish a percentage-by-size distribution; the Semenyih diagram therefore displays evidence and measurement gaps, *not invented particle-size frequencies*. A local research thesis provides important processing-method details. Crucially, *visual-identification accuracy*, *sampling recovery*, *AI segmentation performance*, and *organism prevalence* are different endpoints: absent metrics are reported as not reported, never assumed to be zero or 100%.

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1 Scope, questions and boundaries of inference

The two original questions are refined into three linked, answerable evidence questions. This ensures that a claim about a frequently reported size fraction cannot be confused with a universal environmental distribution or with an instrument detection limit.

1. **RQ1: Freshwater size distribution.** Which particle-size ranges dominate published measurements in river or pond water, and how does the reported dominant fraction depend on the sampling mesh, measurement technique, lower detection limit and size-bin conventions?
2. **RQ1b: Semenyih-specific evidence.** What are the measured microplastic concentration, morphology and size distributions in the Semenyih River *where quantitative size results are actually available*? When size proportions are unavailable in accessible sources, what data must be requested before plotting them?
3. **RQ2: UNM-affiliated methods and verification.** Which records explicitly document a University of Nottingham Malaysia affiliation? Which of these are original water-counting studies rather than reviews or experiments on other matrices? What instrumentation, sample preparation, polymer-confirmation methods, numerical validation and concentration findings are supported by the publication or associated doctoral thesis?

The stakeholder's target is a GPS-referenced 2D surface-water concentration map. In this context, 1–5,000 μm defines a broad *potential MP size domain*, not a guaranteed achievable detection range. Neither a 0.7 μm filter pore nor a 53 μm sieve necessarily sets the minimum optical identification size. Particle length, width, polymer identification, filtration recovery and water volume must be defined individually [1], [2].

2 Dataset provenance and reproducible screening

2.1 Original literature pool and data file

The research group's earlier Web of Science/Scopus workflow identified 18,942 bibliographic records (8,923 WoS and 10,019 Scopus); after removing 7,318 duplicate records and excluding 727 document types, 10,897 were screened. Of these, **10,833 were preliminary include candidates** and 64 required manual review. This is not a completed full-text systematic review.

The uploaded `Bibliometrix_Filtered(1).RData` is a compressed R data workspace (RDX3), byte-identical to an earlier project copy (MD5 `93a3900c5778ad6630281b231d7ae3e9`). In this environment no R interpreter is installed; therefore the reproducible numeric screening was run against the **companion 10,833-row `Include_Candidates.csv` export** available from the same project, not by claiming to have executed R on the workspace. All queries and the row-level results are supplied. The user can reproduce the native R export using the instructions in `README.md`.

2.2 Exact query logic and selection counts

Fields: `TI` (title), `AB` (abstract), and affiliation fields `C1`, `affiliations`, `AU_UN`, `AU1_UN`, `RP`. Matching was case-insensitive. **Size screen**: titles/abstracts containing (i) `microplastic`, (ii) a waterbody term (`river, pond, lake, freshwater, reservoir, stream`, including plurals), and (iii) a size-related expression (`particle size, size distribution, size fraction/class, numeric um/mm, smaller than, less than`). This broad screen deliberately has false positives: merely defining MPs as less than 5 mm is insufficient for inclusion in the final evidence table.

Table 1: Auditable sequential filters of the original 10,833-record pool. Counts are keyword-candidate numbers, not final full-text eligible studies.

Query or step	Records	Interpretation
Initial preliminary candidate corpus	10,833	Previously de-duplicated and document-type screened
Any waterbody word in title/abstract	7,525	Broad freshwater context; may include references rather than study sites
Any size-related expression in title/abstract	3,245	May include only the formal MP definition
Intersection: waterbody + size + microplastic	**2,085**	Further manual study-level screening necessary
Curated studies with reported size-relevant evidence	**17**	Purposeful transparent subset, *not* an exhaustive final synthesis
Any Nottingham affiliation mention	33	Includes UK, Ningbo and ambiguous Nottingham records
Explicit *Nottingham Malaysia* / Semenyih-campus affiliation	**9**	Full UNM-affiliation set under defined metadata rule
Not matching Malaysia-campus rule	24	Retained for audit, not attributed to UNM

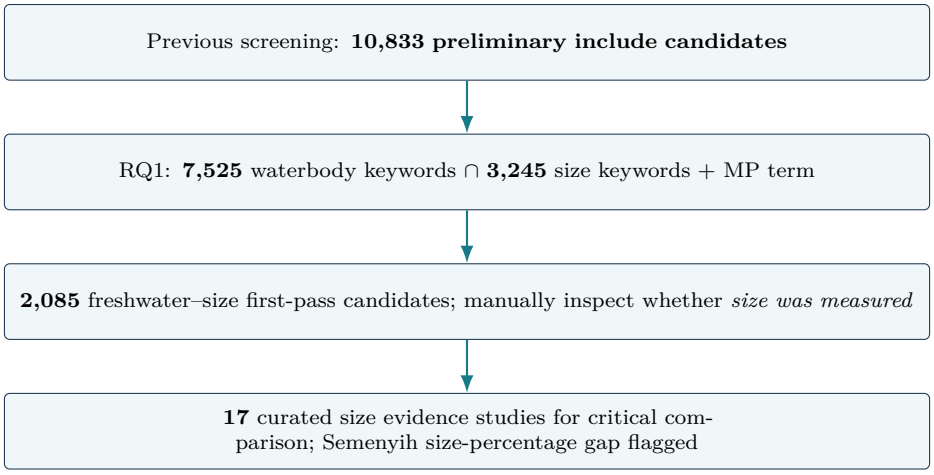


Figure 1: Size-evidence filtering. The 17 examples are a transparent targeted sample, not the total number of eligible studies within 2,085 records.

2.3 How affiliation was filtered

The *Nottingham* query was run across corresponding/author-address and affiliation metadata, but **not** the author names alone. Campus confirmation required an explicit `UNIV NOTTINGHAM MALAYSIA` / `UNIVERSITY OF NOTTINGHAM MALAYSIA`, or an institutional Nottingham address locating the author in Semenyih; merely using a Nottingham email domain or a name such as "University of Nottingham" without Malaysia did not qualify. Each positive record is included in `data/UNM_9_Full_Evidence.csv`, and each other Nottingham result in `data/Nottingham_OtherCampuses_24.csv`. This is *all matches under the named database and rule*, not an exhaustive bibliography of the whole university beyond the database.

2.4 Reproducibility: scripts and paper-level decisions

This Overleaf package contains the actual Python screening code in the `scripts/` subfolder. The original implementation is preserved as `00_original_screening.py`; a revised executable version (`01_screen_candidates.py`) produces a flag and written selection reason for *every one* of the 10,833 corpus records. The companion `curation_size_studies.csv` explicitly separates the *human-reviewed* list of 17 size-evidence examples from the automatic first-pass screen. `03_verify_outputs.py` checks the counts, row identifiers and DOI uniqueness. An RStudio helper can identify and export a matching candidate data frame from the source `.RData`; that helper was supplied for local reproduction, not executed during this analysis.

Table 2: Screening code, input, outputs and permissible inference.

File or stage	Operation	Interpretation
Python first-pass script	Title/abstract keyword intersection for RQ1	Generates 2,085 *potentially relevant* freshwater-size records; not confirmed size reporting.
Human-curated DOI list	Manual DOI-based extraction of 17 studies	An explicitly curated example set, *not* every eligible study among 2,085.
Affiliation query in Python	`C1`, `affiliations`, `AU_UN`, `AU1_UN`, `RP`	Nine strict UNM matches, 24 other/unclear Nottingham records; only within this exported corpus.
Row-level decision audit	Retains all original row IDs and positive/negative flags with reasons	Allows an investigator to review omissions and keyword false positives.
Verification script	Checks membership/count arithmetic	Verifies reproducibility of screening, *not* validity of all study findings.

The source `Include_Candidates.csv` is not embedded in the Overleaf ZIP because it is a large, separately maintained project export. The README explains how to point the scripts to it on Windows. A newer source export with changed records or metadata must be handled as a new screening version, rather than silently mixing counts with this manuscript.

3 RQ1: What microplastic sizes are common in rivers and ponds?

3.1 What comparison across studies can and cannot tell us

A clear pattern in multiple *measured* freshwater studies is that particles smaller than 0.5 or 1 mm are prominent, though the exact leading interval differs substantially by site and method [3], [4], [5], [6], [7]. The opposite is possible: the Songhua River study reports roughly equal contributions from $\leq$1 mm and 1–5 mm classes [8]. There is **no justified global modal size** to quote without harmonising lower detection limits, samplers, polymer verification, fibre measurement conventions, and particle-size definitions. A broader river meta-analysis cautions that sampling methods and missing hydrological context strongly affect apparent riverine MP concentrations [2].

3.2 Representative measured evidence from the screened papers

Table 3: Study-level size findings for freshwater surface water, with mixtures explicitly identified. Values are individual study observations, *not pooled percentages*.

Location/study	Matrix	Reported size evidence	Important interpretation
Langat River, Malaysia [7]	Surface water	$<$ 300: 6.64%; 300–500: 14.19%; 500–1000: 36.81%; 1001–5000 μm: 42.36%	Largest *individual* bin is 1–5 mm; combined $<$ 1 mm is 57.64%.
Selangor River Basin [9]	Surface water	600–1000 μm was the largest reported class, at 26.3%	Regionally relevant; *not* Semenyih.
Cherating River [10]	Aquatic study	0.5–1.0 mm reported prevalent	Malaysia study; full text needed for exact fraction.
Tropical urban ponds, India [6]	Pond surface water	500–1000 μm was the predominant size class	Based on bucket/tube-net samples.
Wei River, China [3]	River water	$<$ 0.5 mm constituted 68.1%	Small fraction dominant in this study.
Lhasa River, China [4]	River surface water	0–0.5 mm constituted 75.83%	Abstract report of river-water class.
Yellow River, Lanzhou [5]	River water	$<$ 1 mm constituted 83.67%	Size and shape jointly discussed.
Vaal River, South Africa [11]	River surface water	0.06–4.95 mm detected; over 89% under 2 mm	Broad range; not same bin system.

Table 3 continued

Location/study	Matrix	Reported size evidence	Interpretation
Vaal River, South Africa [12]	Water and sediment	Over 80% were smaller-than-2-mm fragments/fibres	Combined size-and-shape statement; matrix separation required.
Nhue–Day basin, Vietnam [13]	River water	0.5–2 mm: 71–100% at reported stations	Station-dependent fraction.
Songhua River, China [8]	River water	≤ 1 mm and 1–5 mm: approximately 50% each	Counterexample to universal smaller-size dominance.
Songhua River, China [14]	Surface water	< 0.5 mm and 0.5–1 mm were main classes	No exact combined percentage in abstract.
Wei River Plain [15]	Surface waters	< 0.5 mm dominant	Exact share not in abstract.
Wei River, China [16]	River water	< 500 μm dominant	Consistent regional example.
Seasonal freshwater ponds [17]	Pond surface water	< 1000 μm commonly observed	Pond water; season and sampling matter.
Lake Gahai, China [18]	Lake water	20–50 μm: 64.42%	Highlights micro-scale detections in sensitive methods.
Semenyih River [19]	Water and bed	**No reliable size-class percentages in abstract**	Do not invent a local histogram.

3.3 Semenyih River: the primary local finding

Chen *et al.* reported 160 samples from eight Semenyih River sites (replicates over three dates), with MPs detected in every sample. Mean river-water abundance was **3.12 ± 2.49 particles/L**; bed-surface abundance was **6027.39 ± 16,585.87 particles/m^2**. **Fibres dominated** the observed types; fragments were proportionally more important on the bed than in the water. Variability among replicates at a site was high, and water and riverbed hotspots did not coincide [19]. This is valuable spatial and morphology evidence but **does not establish the predominant Semenyih MP diameter**.

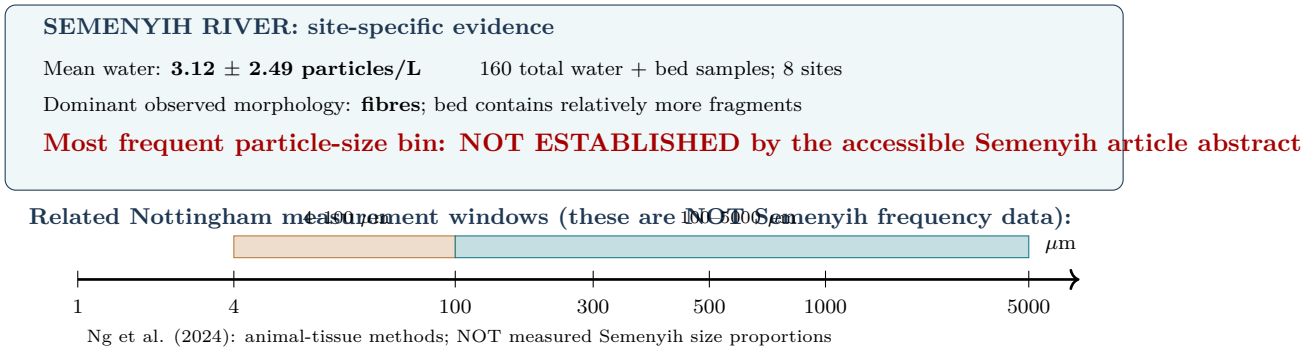


Figure 2: Semenyih-focused evidence diagram. Top: actual local field findings [19]. Bottom: the two *method-capability* windows reported in related Nottingham research [20]; bar length encodes a size range, **not** particle abundance. The actual modal size in Semenyih cannot be inferred.

3.4 Regional size-distribution comparator—Langat is not Semenyih

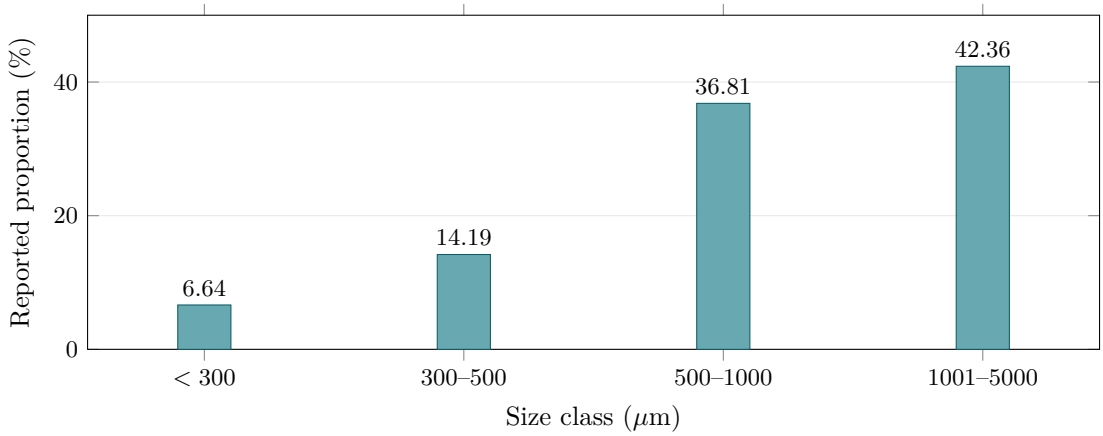


Figure 3: **Langat River comparator ONLY**, reported by Mohd Ali *et al.* [7]. The proportion under 1,000 μm is 57.64%. This is *not* a Semenyih River plot or a pooled Malaysian distribution.

3.5 Implications for selecting a detection size cutoff

For a practical Track A (live boat) or Track B (shoreline processing) device, an initial **100–1,000 μm or 100–5,000 μm** operational target can be considered for imaging-based counting, but would knowingly exclude smaller particles. These are proposed design ranges, not established Semenyih modal sizes. Fine fractions below 100 μm matter scientifically, as illustrated by the 20–50 μm lake measurements, but demand higher-resolution detection and more stringent contamination/verification controls [18], [20]. A robust output must state minimum measurable size, count-versus-mass units and handling losses. Filtration cutoffs and optical resolutions are separate quantities.

4 RQ2: All explicit University of Nottingham Malaysia matches

4.1 Affiliation audit and completeness

Nine records pass the strict campus affiliation rule. They cover **six primary/experimental contributions** and **three reviews or conceptual syntheses** after content appraisal. These are *not* nine laboratory detection-accuracy experiments. A co-author's UNM address is sufficient for inclusion; it does *not* prove that all experimental work was performed on campus or that every listed instrument is owned by UNM. Exact affiliations can be checked in the companion CSV.

Table 4: Every paper with an explicit UNM affiliation in the 10,833 records, classified by its actual purpose.

Year	Paper	Technique / experimental role	Results / detection accuracy
2021	Mofijur *et al.*, sources, distribution and threats [21]	Narrative review; no new UNM sample detection established	No empirical sensor-accuracy rate.
2021	Chen *et al.*, spatio-temporal Langat River [22]	Repeated field sampling, preparation, visual microscopy; subset FTIR confirmation	508 samples; mean 4.39 particles/L; **96% visual identification agreement** in FTIR-checked $n = 85$ subset.
2021	Chen *et al.*, Southeast Asian freshwater review [23]	Literature review of levels, sources and impacts	No new experimental method accuracy.
2022	Lee *et al.*, U-Net counting [24]	Microscope images + U-Net segmentation and post-processing	Human-level counting described; *numerical accuracy not given in abstract.*

Table 4 continued

Year	Paper	Technique / role	Results / accuracy
2022	Chen *et al.*, local riverborne sources [25]	Water sampling, microplastic identification and source apportionment	Road culverts 42.20 ± 35.29 particles/L; reported WWTP removal efficiencies are *not* accuracy.
2023	Chen *et al.*, Semenyih River water/bed [19]	Replicated sampling, preparation and microscopy; bed density separation	3.12 ± 2.49 particles/L water; no per-study detection-accuracy percentage given.
2023	Hanun *et al.*, weathered PET sorption [26]	Controlled PET aging, SEM morphology, FTIR surface chemistry, BET, sorption	Sorption changes measured; *not a river-water MP-counting accuracy study.*
2024	Ng *et al.*, fish and macroinvertebrates [20]	Visual 0.1–5 mm and automated high-resolution 0.004–0.1 mm analysis	95% fish positive, 44% macroinvertebrate positive (*prevalence, not accuracy*).
2026	Alowaisy *et al.*, phytoplankton–plastic interactions [27]	Review/conceptual synthesis of heavy metals and micro/nanoplastics	No new UNM detection-accuracy measurement.

4.2 UNM's documented laboratory workflow: what equipment is evidenced?

Evidence from Chen's doctoral thesis (not a new tenth paper in the 10,833 set). The sampling and processing workflow is described in detail in Chapter 5 [1]:

1. Collect river-water and bed samples and log replicated site/date information.
2. Digest water-sample organic matter using **20 mL of 30% hydrogen peroxide + 20 mL of 0.05 M iron(II) sulfate** (Fenton-type treatment), heated at approximately **60°C for 30 minutes**.
3. Filter water samples using **0.7-μm Whatman GF/F glass-microfibre filters**; oven-dry at approximately **60°C for 24 h**.
4. Visually examine and enumerate particles using a **Nikon SMZ1500 stereoscopic microscope**, approximately **4–180×**; assign morphologies (fibres, fragments, films, beads).
5. For sediment samples, additionally use a **53-μm sieve**, **saturated $ZnCl_2$ density separation**, and **Eppendorf 5810R centrifuge at 10,000 rpm for 10 minutes**, then examine recovered material.

The cited research also reports **FTIR checks of a subset of visual identifications** [22]. These instrument/model details are documented in the project-associated thesis and source paper, but a publication is not a campus asset inventory; verify current instrument access separately. Chemical workflows above require proper safety procedures and equipment and are described as reported methods, *not recommended as an unsupervised shoreline protocol.*

4.3 What does the reported "96% accuracy" actually mean?

The Langat River study states that FTIR checks of **85 visually classified particles** found **96% correct visual identification** [22]. This estimates performance of *visually selected candidates in the checked subset*, conditional on sampling/selection. It is **not** a field-wide sensitivity, independent polymer-confirmation rate for every particle, filtration recovery, or true-negative specificity. It cannot be transferred automatically to a polarising-camera, EIS, or new AI device. The original U-Net study reports human-level image counting, but its searchable abstract does not provide a numeric percent to claim as a validation rate [24].

Table 5: Do not mistake these published percentages for equivalent accuracy metrics.

Source	Number	Meaning
Chen et al. (2021) [22]	**96%** ($n = 85$)	Correct visual identification among a checked set using FTIR; *not* sampling recovery.
Ng et al. (2024) [20]	95% fish; 44% macroinvertebrates	Organisms in which MPs were detected by the low-resolution method; *not* detection sensitivity.
Chen et al. (2022) [25]	30.95–69.94%	Wastewater-treatment-plant MP removal efficiencies; *not* analytical accuracy.
Chen et al. (2023) [19]	160/160 positive samples	Prevalence across field samples; *not* sensitivity, specificity or recovery.
Lee et al. (2022) [24]	No abstract-level percentage	Qualitative "human-level" counting description; require full paper/test set for precision, recall and error.

5 Critical synthesis and implications for the project

5.1 What was actually learned from the local research?

Semenyih water and bed need separate consideration: neither the local samples nor the spatial maxima are interchangeable. Repeated stations and adequate sampled volumes are important for georeferenced maps [19]. **Langat shows temporal change:** all 508 river-water samples contained MPs; abundance was 4.39 particles/L on average and much higher in some tributaries, with a positive relation to river discharge and sediment/turbidity indicators [22]. **Sources matter:** road drainage was a larger local source than residential or industrial culverts in the source-apportionment study [25]. **Size-resolution matters:** 2024 biological monitoring found much higher counts when the high-resolution method was used; this cannot be interpreted as a measured river-water size distribution [20]. **Imaging is feasible but not chemically specific:** the UNM U-Net contribution accelerates counts on microscope images, whereas chemical confirmation is a separate test [22], [24].

5.2 Track A versus Track B for a 2D map

Table 6: Method implications based on the findings and the restriction on routine specialist laboratory access.

Issue	Track A: live measurement	Track B: collected sample, shore processing
Size threshold	Flow-cell path and optics impose minimum size; must state it	Filter/pore size and microscope resolution each need calibration
Concentration at 3.12 particles/L	Low counts require adequate observed water volume and longer averaging	Larger known-volume samples can be collected and analysed in batches
Small particles	Especially difficult at high flow in complex water	Portable enrichment/microscopy may help, but below 100 μm challenging
UNM comparison	U-Net concept potentially adaptable but does not solve polymer selectivity	Local microscopy/FTIR studies offer precedent for a controlled reference procedure
Evidence standard	Particle count/L + known sensor volume + false-positive testing	Particle count/L + recovered fraction + blanks; optional limited external polymer ID
Map output	Continuous or segmented estimates, with latency and QC labels	Discrete geotagged observations, followed by interpolation with held-out checks

5.3 Recommended next evidence request

To produce an *actual Semenyih percentage-by-size histogram*, obtain the corresponding particle-level dataset or full size-class table (if collected) from Chen et al. (2023) or the authors' thesis attachments. The published source indicates that the corresponding-author dataset is available on reasonable request [19]. Obtain the original measured particle dimensions, morphology, sample size, minimum measurable size, and polymer-confirmation status; re-bin only after obtaining them. Do not substitute Langat or other Malaysian river fractions. If the team generates new data, retain raw images, calibration scales, size-bin definitions, blanks, volume, sampling date, and GPS coordinates.

6 Study limitations and evidence-status statement

The 17 size papers are a reproducible *curated example set* drawn from 2,085 preliminary text matches, **not exhaustive final full-text inclusion**. The nine UNM papers are exhaustive under the explicit affiliation rule *within this specific 10,833-record corpus*. University-owned laboratory inventory, transfer of methods to a new sensor, unreported detection efficiencies, and original Semenyih size frequencies cannot be inferred from abstracts. Any claims taken from the source thesis or publisher full text are distinguished from the corpus-wide title/abstract screening. Direct comparisons among studies require harmonised sampling method, minimum size, sample matrix, counting units, polymer identification and quality assurance [2].

IEEE-style references with DOI links

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